Creative

A1 ASSIGNMENT

Interface Design

04_Shweta - Team 8
Jingyi Ao Yiyang Li
jiao0884 yili0658
IDEA 9105

LIST OF CONTENTS

01 BACKGROUND RESEARCH

02 ONLINE ETHNOGRAPHY METHOD

03 PERSONAS

04 JOBS TO BE DONE

05 USER JOURNEY

06 ENVISIONED SOLUTION + DEVICE ALLOCATION

BACKGROUND RESEARCH

What Is Karaoke?

- Singing with instrumental tracks and on-screen lyrics
- More than entertainment: a participatory social experience
- Common in public and semi-public spaces (Encyclopaedia Britannica, 2026)

Origin and Development

- Encourages bonding, participation, and performance
- Also involves vulnerability, embarrassment, and anxiety
- Karaoke is both enjoyable and emotionally complex (Bryan & Tanenbaum, 2019; Leung et al., 1998)

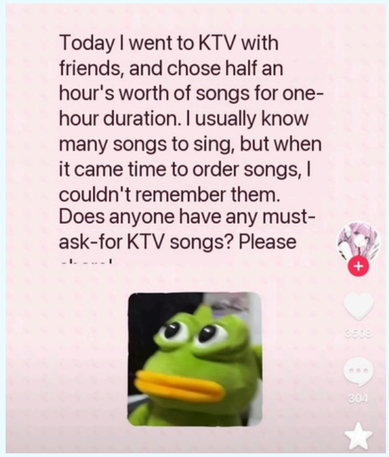
Origin and Development

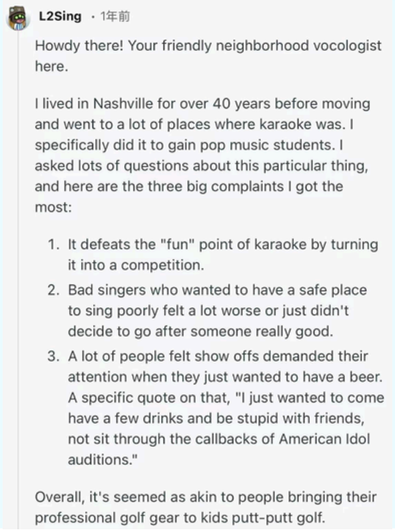
- Originated in Japan in the early 1970s
- Expanded globally over time
- Evolved into digital, interactive systems (Encyclopaedia Britannica, 2026; Swaminathan, 2009)

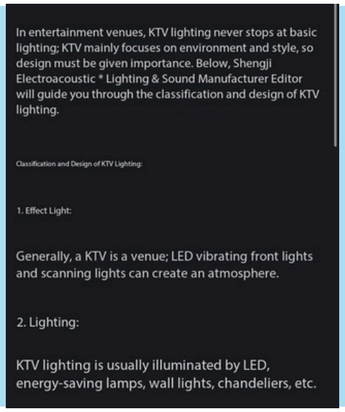
Design Opportunity

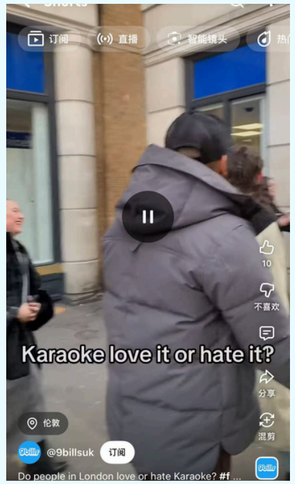
- Karaoke now relies on connected digital interfeces
- Technology shapes both interaction and participation
- Anxiety and social pressure create strong design opportunities (Shin, 2023; Dobos et al., 2018)

ONLINE ETHNOGRAPHY METHOD

CONTEXT	DATA RECORD	RECORDED OBSERVATIONS	INTERPRETATIONS	THEME
Platform: Douyin Date: November 17, 2025 Topic: Difficulty choosing songs during karaoke Area: song selection/memory/decision-making		• Users say they know many songs, but cannot recall them when placing an order. • They sound slightly frustrated, but are still hoping for assistence. • This suggests that users rely more on their own memory than on the system's assistance features.	• When time is of the essence, users may find the task of selecting songs too mentally taxing. • Karaoke systems may not be able to effectively support song search or recommendation functions.	Song selection difficulty / memory recall issues

CONTEXT	DATA RECORD	RECORDED OBSERVATIONS	INTERPRETATIONS	THEME
• Platform: Reddit • Date: 1 year ago • Topic: A self-identified vocologist summarises common complaints about karaoke becoming too competitive • Relevant research area: social atmosphere, competition, safety, casual participation		• People often say that once karaoke turns into a competition, it ceases to be fun. • Some users hope that karaoke will remain a ‘safe space where one can sing as badly as one likes without worry’.	• Many users view karaoke as a relaxed and enjoyable form of entertainment rather than a stage for showcasing their singing talents. • Competition and comparison may undermine inclusivity and reduce the willingness to participate.	Social pressure / competition versus safe participation

CONTEXT	DATA RECORD	RECORDED OBSERVATIONS	INTERPRETATIONS	THEME
• Platform: Zhihu • Date: 2021-03-29 • Topic: a post about KTV lighting design and how atmosphere is central to KTV spaces • Relevant research area: lighting, ambience, and environmental design in karaoke spaces		• The post clearly states that atmosphere is a key element of the KTV experience. • It points out that lighting serves not only a practical purpose but is also used to create atmosphare and style.	• The karaoke experience depends not only on the songs or the interface, but also, to a certain extent, on the atmosphere. • Lighting has a significant impact on the sense of immersion, excitement or comfort within a karaoke room.	Lighting and atmosphere as part of karaoke experience
• Platform: Bilibili • Date: 2025-08-11 • Topic: a user feels embarrassed singing because their voice blends into the original track • Relevant research area: singing confidence, self-consciousness, audio feedback, participation anxiety in karaoke		• This user is concerned about both pitch accuracy and the quality of their voice. • They deliberately lower their voice so as not to be heard clearly. • This comment highlights the disconnect between a willingness to participate and confidence in performing.	• Karaoke does not always instil confidence in performers; for some users, it can actually trigger feelings of self-doubt and anxiety. • This presents to focus on boosting confidence, providing vocal guidance and offering adjustable backing track features..	Sing anxiety / low-confidence participation

CONTEXT	DATA RECORD	RECORDED OBSERVATIONS	INTERPRETATIONS	THEME
• Platform: Red Note • Date: 11/09/2025 • Topic: A user discusses the possibility of combining KTV singing with fitness or exercise-related content • Relevant research area: alternative karaoke experiences	Isn't singing in KTV a business opportunity? Now so many people go to KTV singing to lose weight cult cultivation, why not KTV to match the fitness of the mv!! Open at least one column with a rehearsal version of 100 hits. Anyway, there are some songs MV is also randomly matched, when it comes to copyright, there is no copyright (song copyright should be purchased in the whole package?)🤣Who can run a KTV like this?! I'm willing to pay for it!!!	• The user believes that singing in a karaoke room may aid weight loss and fitness, and has suggested combining songs with music videos featuring exercise routines. • The user has indicated a willingness to pay for this.	• Some users believe that karaoke is experience that can be dynamic and physically engaging. • This suggests that we have an opportunity to redesign the karaoke interface to support physical movements, instructional guidance.	Karaoke as a physically engaging and health-related activity
• Platform: YouTube Shorts/street interview context • Date: 1 month ago • Topic: public attitudes toward whether people love or hate karaoke • Relevant research area: public perception, attitude diversity, social acceptance of karaoke		• This question assumes that users may have strong emotional reactions to karaoke, whether positive or negative. • Karaoke is regarded as a topic of public culture, and discussions about it can extend beyond the venue itself. • The data suggests that karaoke holds different meanings for different people.	• Some users may prefer group singing, whilst others wish to maintain the privacy of their personal singing space. • This suggests that karaoke systems and environments should better support clear role differentiation, invitations for duets, or visualised round-singing mechnisms.	Singing over others / blurred boundaries in group participation

context dependence / cultural and situational variation in karaoke experience

Venue hygiene and physical discomfort

social pressure / competition versus safe participation

Group dynamics / karaoke as collaborative social interaction

Song selection difficulty / memory recall issues

Vocal strain / mismatch between song difficulty and user ability

KTV as a multi-purpose social environment

Loss of physical interaction in digital karaoke

Song library dependence / emotional impact of losing preferred songs

Embodied participation / following the MV through movement

Lighting and atmosphere as part of karaoke experience

Queue domination and emotional exclusion

Social discomfort/fear of singing among friends

Environmental comfort

Alternative participation / listener-type users

Monetised gamification and unfair scoring

erformance inconsistency / sound environment issues

Negative environmental experience

Performance anxiety / fear of singing out of tune

collaborative performance

Cost and affordability of karaoke

Positive solo karaoke experience

Audio feedback / lack of singing support

Performance anxiety

dislike of karaoke environment

Audio feedback / self-perception and vocal control

DIY digital karaoke / platform workaround

Confidence-building strategies / supportive song choice

Solo karaoke safety concerns and socially anxious participation

Solo karaoke / private immersive participation

Price complexity and participation barriers

Karaoke etiquette and shared social boundaries

karaoke privacy and interruption

Digital usability / karaoke search and content discovery

Karaoke as a physically engaging and health-related activity

Emotional support / confidence-building in karaoke participation

Performance anxiety / fear of judgement

Context fit / karaoke depends on venue and audience culture

Alternative participation

Different KTV types and user contexts

KTV as an integrated environmental system

Solo karaoke as comfort and autonomy

Social awkwardness in group decision-making / hidden participation barriers

Social visibility and spatial anxiety

Song ownership / tension between solo performance and group participation

Queue unfairness / microphone domination / exclusion in group

Singing anxiety / low-confidence participation

Environmental immersion / value of venue-based karaoke

Fragmented music taste / loss of shared group resonance

KTV as a photogenic self-presentation space

Service breakdown, loss of control, and privacy anxiety

KTV as mainstream group entertainment

Solo karaoke as relief from group pressure

First-time anxiety / entry barriers to solo karaoke

Karaoke as companionship / relational bonding

Low-pressure participation / fun over technical skill

Duet karaoke / shared participation as attraction

Polarised attitudes toward karaoke

Singing over others / blurred boundaries in group participation

Systemic service failure and comfort breakdown

Performance Anxiety

Performance anxiety

Performance anxiety / fear of judgement

Singing anxiety / low-confidence participation

Social discomfort/fear of singing among friends

Performance anxiety / fear of singing out of tune

First-time anxiety / entry barriers to solo karaoke

Emotional support / confidence-building in karaoke participation

Social Participation

social pressure / competition versus safe participation

Group dynamics / karaoke as collaborative social interaction

KTV as a multi-purpose social environment

Embodied participation / following the MV through movement

Alternative participation / listener-type users

collaborative performance

Karaoke as a physically engaging and health-related activity

Alternative participation

Song ownership / tension between solo performance and group participation

Queue unfairness / microphone domination / exclusion in group

Karaoke as companionship / relational bonding

Low-pressure participation / fun over technical skill

Duet karaoke / shared participation as attraction

Polarised attitudes toward karaoke

Singing over others / blurred boundaries in group participation

Queue domination and emotional exclusion

Karaoke etiquette and shared social boundaries

KTV as mainstream group entertainment

Solo karaoke as relief from group pressur

Social awkwardness in group decision-making / hidden participation barriers

Environmental Comfort

Environmental comfort

Service breakdown, loss of control, and privacy anxiety

Negative environmental experience

dislike of karaoke environment

Solo karaoke safety concerns and socially anxious participation

Solo karaoke / private immersive participation

karaoke privacy and interruption

Context fit / karaoke depends on venue and audience culture

Different KTV types and user contexts

Social visibility and spatial anxiety

Environmental immersion / value of venue-based karaoke

KTV as an integrated environmental system

Lighting and atmosphere as part of karaoke experience

Venue hygiene and physical discomfort

Positive solo karaoke experience

Solo karaoke as comfort and autonomy

KTV as a photogenic self-presentation space

context dependence / cultural and situational variation in karaoke experience

Song Selection

Song selection difficulty / memory recall issues

Vocal strain / mismatch between song difficulty and user ability

Song library dependence / emotional impact of losing preferred songs

Confidence-building strategies / supportive song choice

Fragmented music taste / loss of shared group resonance

System Support

Loss of physical interaction in digital karaoke

performance inconsistency / sound environment issues

Audio feedback / lack of singing support

Audio feedback / self-perception and vocal control

DIY digital karaoke / platform workaround

Digital usability / karaoke search and content discovery

Systemic service failure and comfort breakdown

Cost and affordability of karaoke

Price complexity and participation barriers

Monetised gamification and unfair scoring

Performance Anxiety

Performance anxiety

Performance anxiety / fear of judgement

[illegible] participation

Social discomfort/fear of singing among friends

Performance anxiety / fear of singing out of tune

First-time anxiety / entry barriers to solo karaoke

Emotional support / confidence-building in karaoke participation

Environmental Comfort

Environmental comfort

Service breakdown, loss of control, and privacy anxiety

Negative environmental experience

dislike of karaoke environment

Solo karaoke safety concerns and socially anxious participation

Solo karaoke / private immersive participation

karaoke privacy and interruption

Context fit / karaoke depends on venue and audience culture

Different KTV types and user contexts

Social visibility and spatial anxiety

Environmental immersion / value of venue-based karaoke

KTV as an integrated environmental system

Lighting and atmosphere as part of karaoke experience

Venue hygiene and physical discomfort

Positive solo karaoke experience

Solo karaoke as comfort and autonomy

KTV as a photogenic self-presentation space

context dependence / cultural and situational variation in karaoke experience

Song Selection

Song selection difficulty / memory recall issues

Vocal strain / mismatch between song difficulty and user ability

Song library dependence / emotional impact of losing preferred songs

Confidence-building strategies / supportive song choice

Fragmented music taste / loss of shared group resonance

System Support

Loss of physical interaction in digital karaoke

performance inconsistency / sound environment issues

Audio feedback / lack of singing support

Audio feedback / self-perception and vocal control

DIY digital karaoke / platform workaround

Digital usability / karaoke search and content discovery

Systemic service failure and comfort breakdown

Cost and affordability of karaoke

Price complexity and participation barriers

Monetised gamification and unfair scoring

Social Participation

social pressure / competition versus safe participation

Group dynamics / karaoke as collaborative social interaction

KTV as a multi-purpose social environment

Embodied participation / following the MV through movement

Alternative participation / listener-type users

collaborative performance

Karaoke as a physically engaging and health-related activity

Alternative participation

Song ownership / tension between solo performance and group participation

Queue unfairness / microphone domination / exclusion in group

Karaoke as companionship / relational bonding

Low-pressure participation / fun over technical skill

Duet karaoke / shared participation as attraction

Polarised attitudes toward karaoke

Singing over others / blurred boundaries in group participation

Queue domination and emotional exclusion

Karaoke etiquette and shared social boundaries

KTV as mainstream group entertainment

Solo karaoke as relief from group pressur

Social awkwardness in group decision-making / hidden participation barriers

Theme

- **Performance Anxiety**
- Song Selection
- Environmental Comfort
- System Support
- Social Participation

Insights:

- The reason many people avoid karaoke is not that they dislike singing, but because their anxiety makes them feel that take part carries a high risk and will expose their emotions.
- Anxiety about singing karaoke is influenced not only by an individual's self-confidence but also by the social context, including who is present, how others behave, and whether the atmosphere is supportive or judgmental.

PERSONAS

Basic information



Mia

"The Anxious Participant"

- Age 21
- Occupation University student
- Location Sydney
- Goes to karaoke with friends
- Prefers low-pressure participation

Bio

Mia likes the social atmosphere of karaoke, but she often experiences performance anxiety. She is afraid of singing out of tune, being compared with stronger singers, or becoming awkward in front of friends. She prefers joining group choruses or singing quietly with background vocals rather than performing alone. For Mia, a good karaoke experience is one where she can participate without feeling exposed or judged.

Goal | Frustration | Need

Personality

Introvert	Extrovert
Analytical	Creative
Busy	Time rich
Messy	Organized
Independent	Team player
Passive	Active
Safe	Risky

Interests

- Karaoke with friends
- Group choruses
- Familiar songs

Influences

- Fear of judgement
- Stronger singers in the room
- Social pressure from friends

Goals

- Sing with more confidence
- Enjoy karaoke comfortably
- Join without feeling exposed

Needs and expectations

- Low-pressure participation
- Supportive group atmosphere
- Safer ways to join singing

Motivations

- Social belonging
- Enjoyment of music
- Comfortable participation

Pain points and frustrations

- Singing out of tune
- Being compared with others
- Solo performance anxiety

PERSONAS



John

The Overthinking Song Planner

- Age：24
- Occupation：Graduate employee
- Location：Sydney
- More info：Usually joins karaoke after dinners, birthday gatherings, or work socials
- More info： Likes music, but becomes anxious when he has to choose songs in front of others
- More info：Often spends too much time thinking about what song is "safe" to sing

Bio

Ethan enjoys the idea of karaoke and wants to take part when friends or colleagues organise a KTV session. However, his anxiety starts before he even sings. He worries about choosing the wrong song, picking something too difficult, or making the mood awkward if the song does not fit the room. Even when he knows many songs, he often cannot remember them in the moment. He overthinks the order of songs, whether people know the song, and whether he can finish it without mistakes. As a result, he often delays his turn, chooses very familiar songs, or lets others go first.

Goals | Motivations | Frustrations

Personality

Introvert	Extrovert
Analytical	Creative
Busy	Time rich
Messy	Organized
Independent	Team player
Passive	Active
Safe	Risky

Interests

- Pop and nostalgic songs
- Group dinners and casual hangouts
- Short-form music clips and karaoke recommendations

Influences

- Group mood and audience response
- Whether other people know the song
- Previous embarrassing karaoke experiences

Goals

- Choose songs that feel safe and appropriate
- Join karaoke without making the room awkward
- Feel prepared before it is his turn to sing

Needs and expectations

- Quick access to familiar and suitable songs
- Guidance on song difficulty and audience fit
- A more supportive and less rushed song-selection process

Motivations

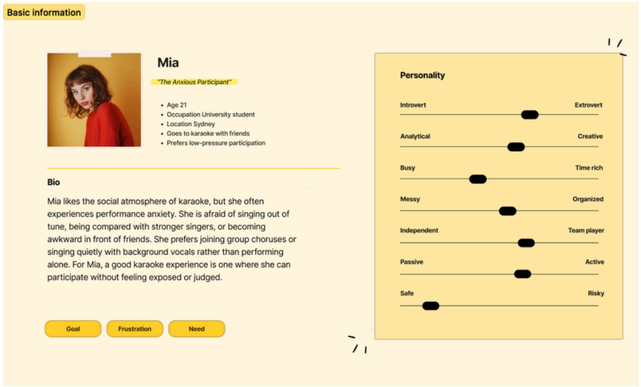
- Wants to stay included in social events
- Enjoys singing when he feels prepared
- Wants to avoid embarrassment while still participating

Pain points and frustrations

- Cannot remember songs when it is time to order
- Worries about picking songs that are too hard or too slow
- Feels pressure when others are waiting for him to decide

JOBS TO BE DONE

Situation	Motivation	Expected outcome
When: I join karaoke with friends	I want to: I want to sing without feeling judged	so I can: so I can enjoy karaoke more confidently
When: I am in a room with stronger singers	I want to: I want to avoid feeling compared or embarrassed	so I can: so I can still participate without becoming anxious
When: It becomes my turn to sing alone	I want to: I want a lower-pressure way to join in	so I can: so I can participate without feeling too exposed
When: I feel nervous in karaoke settings	I want to: I want the environment to feel supportive and safe	so I can: so I can relax and stay involved in the group

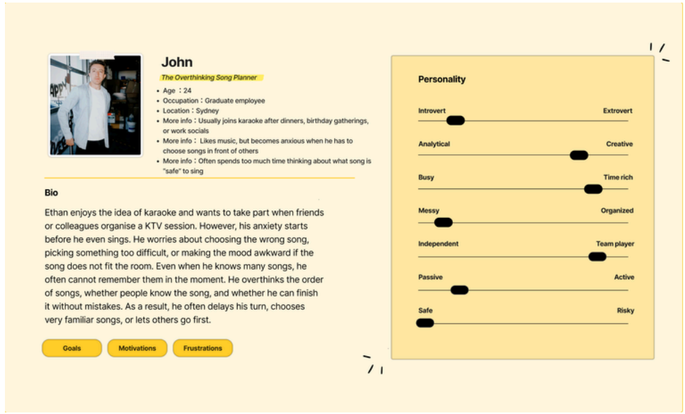


- Based on our personas and ethnographic findings, we identified the following key jobs users want to accomplish.

Importance (1-10)	Satisfaction of outcom...	Opportunity
9	3	9 - 3 = 6
9	4	9 - 4 = 5
8	3	8 - 3 = 5
8	4	8 - 4 = 4

JOBS TO BE DONE

	Situation	Motivation	Expected outcome
When	I am about to choose a karaoke song in front of others	I want to feel confident in my choice	so I can participate without overthinking or feeling embarrassed
When	I join a group karaoke session	I want to choose a song that suits both my ability and the group mood	so I can enjoy the experience without making the situation awkward
When	It is almost my turn to sing	I want to feel prepared for what I am about to perform	so I can reduce anxiety and stay socially engaged
When	other people are waiting for me to pick a song	I want to make a decision quickly without second-guessing myself	so I can stay involved without feeling pressured or awkward



- Based on our personas and ethnographic findings, we identified the following key jobs users want to accomplish.

Importance (1-10)	Satisfaction of outcomes ...	Opportunity
9	4	9 - 4 = 5
9	3	9 - 3 = 6
8	4	8 - 4 = 4
8	5	8 - 5 = 3

USER JOURNEY

Anxiety Journey in Karaoke Participation

This journey shows how anxiety develop across karaoke participation, and how different touchpoint can reduce pressure and uncertainty

1.Before going



Touchpoint: smartwatch

The user starts worrying after being invited to karaoke.
A smartwatch helps them prepare a few familiar songs.

2.Arriving at Karaoke



Touchpoint: Tablet

Watching others sing can increase anxiety.
A tablet helps reduce uncertainty by showing the queue and current songs.

3.Choosing a song



Touchpoint: Tablet

Choosing a song in front of others creates pressure.
A smartwatch and tablet help the user choose more quickly.

4.Singing and after singing



Touchpoint: Tablet+Smartwatch

The user focuses on avoiding mistakes while singing.
With more support, they feel more confident and willing to join again.

ENVISIONED SOLUTION + DEVICE ALLOCATION

Our envisioned solution is a low-pressure karaoke support system that responds to anxiety across the user journey. This direction was chosen because our user journey showed that anxiety begins before arrival, increases during song selection, and continues during participation. By combin smartwatch and tablet interactions, the concept aims to suppoet preparation before arrival, safer song selection, and lower-pressure participation during the session.

Device roles

Smartwatch

Supports pre-visit preparation and emotional support

- save familiar songs before arriving
- reduce uncertainty before the session
- help users feel more prepared and less anxious

Device roles

Tablet

Supports in-room guidance and lower-pressure participation

- show the queue and current songs clearly
- support safer song selection
- reduce pressure during participation

Device allocation

- Jingyi Ao – **Tablet flow**

Intervenes in the “arriving at KTV,” “choosing a song,” and “participating” stages by supporting in-room guidance, safer song selection, and lower-pressure participation.

- Yiyang Li – **Smartwatch flow**

Intervenes in the “before going” stage by supporting preparation before arrival and reducing anticipatory anxiety.

References

Britannica Editors. (2026, February 16). *Karaoke. Encyclopaedia Britannica*. https://www.britannica.com/art/karaoke

Bryan, J. S., & Tanenbaum, T. J. (2019). *Adapting the empty orchestra: The performance of play in karaoke. DiGRA Digital Library*. https://dl.digra.org/index.php/dl/article/download/1114/1114/1111

Dobos, B., Piko, B. F., & Kenny, D. T. (2019). *Music performance anxiety and its relationship with social phobia and dimensions of perfectionism. Research Studies in Music Education, 41*(3), 310–326. https://doi.org/10.1177/1321103X18804295

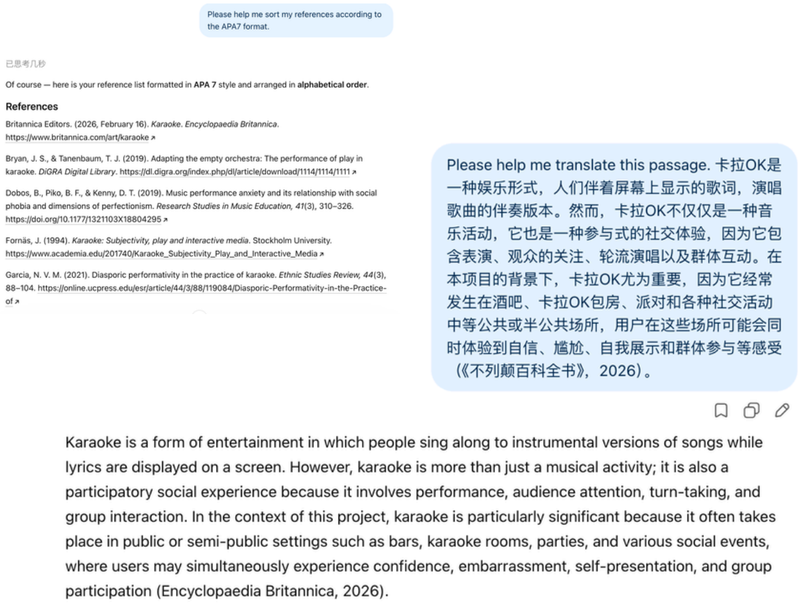
Leung, C. M., Lee, G., Cheung, B., Kwong, E., Wing, Y. K., Kan, C. S., & Lau, J. (1998). *Karaoke therapy in the rehabilitation of mental patients. Singapore Medical Journal, 39*(4), 166–168. https://pubmed.ncbi.nlm.nih.gov/9676147/

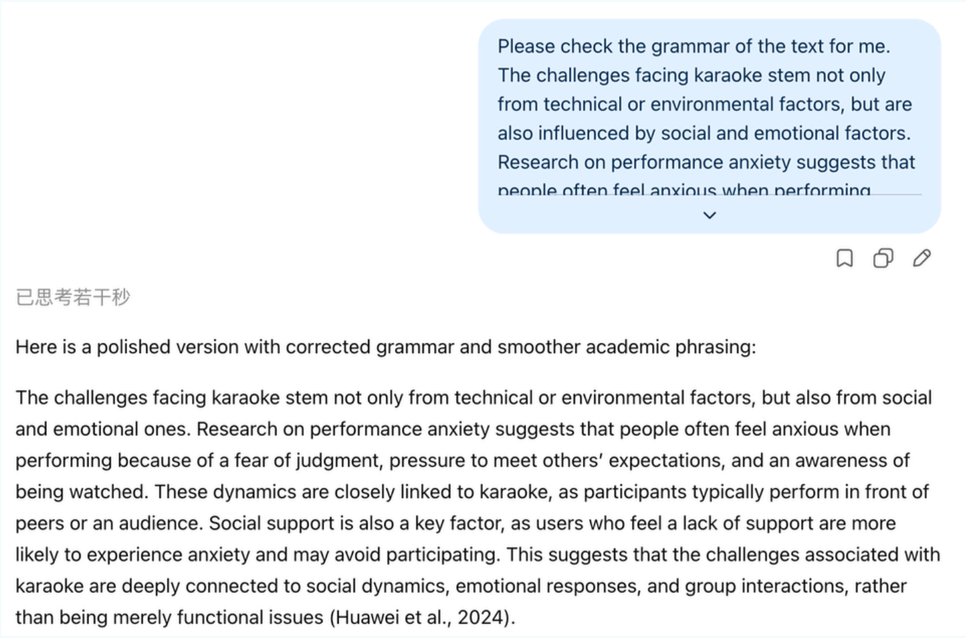
Shin, C. (2023). *Design and implementation of impromptu mobile social karaoke for digital cultural spaces in the new normal era. Applied Sciences, 13*(22), Article 12319. https://doi.org/10.3390/app132212319

Swaminathan, A. (2009, February 25). *The father of karaoke. World Intellectual Property Organization*. https://www.wipo.int/en/web/wipo-magazine/articles/the-father-of-karaoke-36671

AI Acknowledgment

We acknowledge the use of ChatGPT for grammar support, written expression. We also used a Canva template to support the visual presentation of this report. All core content, including the user journey stages, pain points, insights, online ethnography, personas, and jobs-to-be-done analysis, was developed by our group based on our own research. AI was not used to generate primary data.





THANKS FOR WATCHING